# TARAKSHA

## SMART FLOOD RESCUE & RELIEF DELIVERY BOAT

### 1. TITLE

TARAKSHA – Smart Flood Rescue & Relief Delivery Boat

### 2. METHODOLOGY

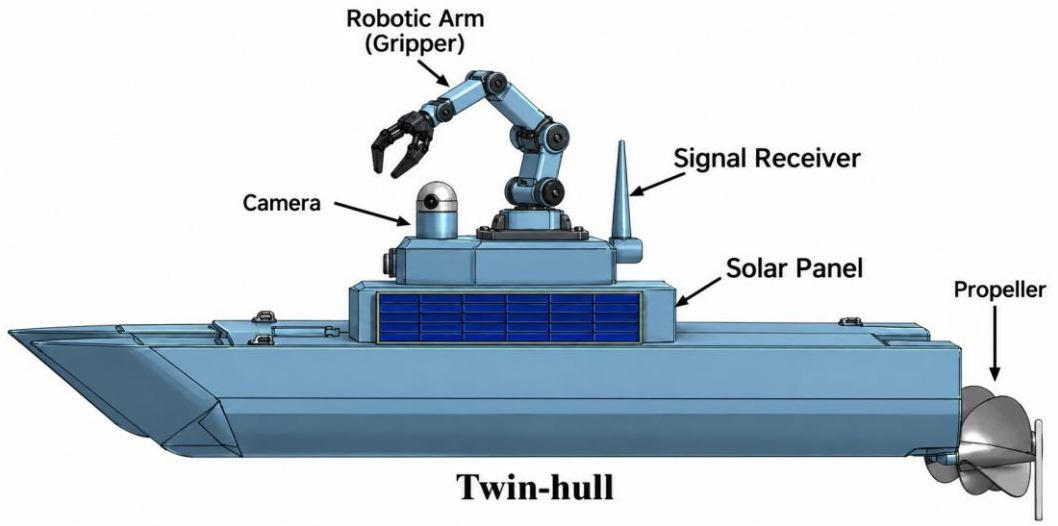
TARAKSHA is a remotely operated twin-hull flood-response boat designed to support trained rescue teams during flood situations by inspecting flooded locations and delivering small emergency supplies. An ESP32 acts as the main controller, while a 4G router with SIM provides primary internet communication where cellular coverage is available. An independent RC transmitter and receiver provide backup control. A 360° camera enables live visual monitoring, while GPS, compass and IMU provide location, heading and movement information. Ultrasonic and sonar sensing assists with nearby above and below water obstacle awareness. Battery and water-leak monitoring support safer operation. A waterproof payload compartment carries food, drinking water, first-aid items and other small relief materials, with a remote-release mechanism for delivery. The system also incorporates a GPS-based Return-to-Home (RTH) concept. A robotic arm in the proposed design can support handling or positioning small relief packages.

### 3. PROPOSED BLOCK DIAGRAM

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Mobile/ RF Transmitter | → | 4G Network/ 2.4/5 ghz | → | 4G Router& RX TX Transmitter | → | 2 way communication | → | Backup communication |
| 360° Night Vision Camera+ ultrasonic + sonar | → | Visual/obstacle detection | → | Wifi cam+ Ultrasonic + sonar | → | Live Monitoring in above and below water | → | Mission Data |
| Robotic Arm | → | Relief Handling | → | Mechanical arm | → | Controlled Rescue Support | → | Help to save human |
| GPS + RTH | → | Return Home | → | RTH | → | Report Status | → | Automatically Return to Home |

### 4. PROPOSED OUTCOME

The proposed outcome is a compact rescue-support platform capable of remote navigation, live visual monitoring, location tracking, basic obstacle awareness and delivery of limited emergency supplies. TARAKSHA is intended to help reduce unnecessary exposure of rescue personnel during initial inspection and relief delivery. Its modular design allows future upgrades such as night vision, thermal imaging, stronger communication and improved navigation. The robotic arm can further support the handling or positioning of small relief packages. The system is intended as a support tool for trained rescue personnel.




TARAKSHA CAD MODEL • FLOOD-RESCUE PLATFORM

| CONTROL | CORE SYSTEM | DEVELOPMENT COST |
|---|---|---|
| 4G + RC BACKUP | ESP32 + CAMERA + GPS | ₹55,000-65,000 |

LAUNCH → CONNECT → NAVIGATE → OBSERVE → DETECT → HUMAN VERIFY → DELIVER / ASSIST → REPORT → RETURN